

ALPACA AI TRADING AGENTS HACKATHON · 2026

Capitol Desk

Faaris Ahmed

An autonomous options desk that reads congressional financial disclosures, decides whether the signal is still alive, and trades it — end to end, unattended.

capitol-desk.onrender.com · Claude Opus 5 · Alpaca MCP Server · Alpaca Trading API

Congress trades. The law says they must tell us.

The STOCK Act requires every member of Congress to publicly disclose any transaction over \$1,000 — within 45 days. Those filings are free, public, and almost unreadable.

368

Periodic Transaction Reports filed in 2026

31.5 days

Median lag from trade to disclosure

12%

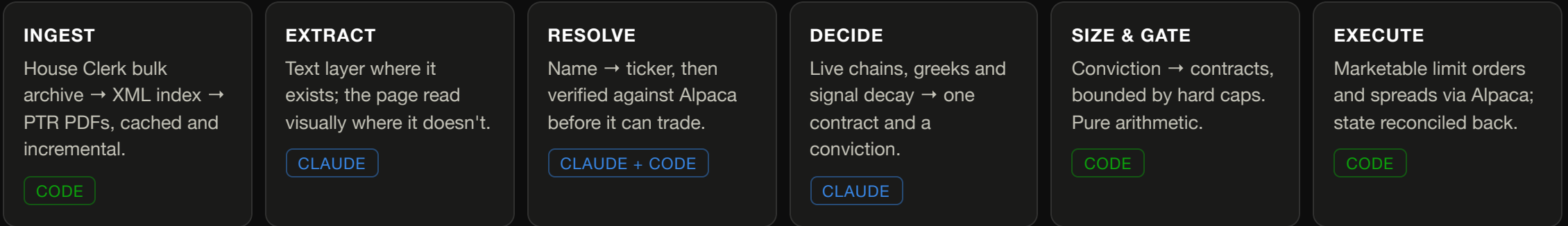
Filed as scanned paper — zero machine-readable text

Public filings. Trading on them is not insider trading — it is the opposite: acting on information the law requires be published.

- Our answer:** Claude reads the text layer where one exists, and reads the **page visually** where one doesn't. On that 274-row scan it extracted every transaction — and correctly ignored the example row.

03

Six stages, and the model only touches two



Claude decides *what* to trade. Code decides *how much*.

Position sizing and risk limits are plain arithmetic in a config file and a gate function. There is no prompt a model can write that gets it a bigger position.

A real filing, a real order



PERIODIC TRANSACTION REPORT

Filing ID #20035143

Clerk of the House of Representatives • Legislative Resource Center • B81 Cannon Building • Washington, DC 20515

FILER INFORMATION

Name: Hon. Nancy Pelosi
Status: Member
State/District: CA11

TRANSACTIONS

ID	Owner	Asset	Transaction Type	Date	Notification Date	Amount	Cap. Gains > \$200?
	SP	Bloom Energy Corporation Class A Common Stock (BE) [ST] FILING STATUS: New DESCRIPTION: Purchased 10,000 shares.	P	07/24/2026	07/24/2026	\$1,000,001 - \$5,000,000	☐
	SP	Bloom Energy Corporation Class A Common Stock (BE) [OP] FILING STATUS: New DESCRIPTION: Purchased 100 call options with a strike price of \$100 and an expiration date of 6/17/27.	P	07/24/2026	07/24/2026	\$1,000,001 - \$5,000,000	☐
	SP	Bloom Energy Corporation Class A Common Stock (BE) [ST] FILING STATUS: New DESCRIPTION: Purchased 5,000 shares.	P	07/28/2026	07/28/2026	\$500,001 - \$1,000,000	☐
	SP	Bloom Energy Corporation Class A	P	07/28/2026	07/28/2026	\$500,001 -	☐

WHAT THE FILING SAYS

Pelosi, filed 21 Aug. Bloom Energy calls, **\$100 strike, expiring 17 Jun 2027** — a deep in-the-money LEAP used as stock replacement.

WHAT THE DESK FOUND

BE had already run **+13.9%** since her trade date. The outright June-2027 contract now costs **\$8,116** against a **\$5,000** per-trade cap — and every affordable strike was 100% time value at 88% implied vol: **volatility risk the filer never took.**

WHAT IT DID

Built a vertical debit spread — long the 175 call (delta 0.75), short the 230. Same expiry, same thesis, **\$2,071 of risk instead of \$8,116**. Conviction cut to 0.45 because the move had partly happened.

It declines far more than it takes

650

Transactions parsed

635

Passed over

3

Trades placed

DECLINED ON THE MERITS — VERBATIM

PG — "a \$1k–\$15k purchase inside a 401(k) subholding of a family trust: a routine retirement-account allocation into a mega-cap consumer staple, not an informed, concentrated bet ... there's no thesis to express."

AMAT — "the quote set is internally incoherent: every listed call is priced above the maximum theoretical value of a call ... there is no way to size or price an honest replication." (It had caught a bug in our own price feed.)

Anyone can build a bot that copies Pelosi. The engineering is in the 635 it refused.

WE MEASURED IT

Copying Congress indiscriminately does not work

Event study over 143 filings and 826 disclosed purchases. Entry is the **filing** date, not the transaction date — a trade is private for up to 45 days, so entering earlier is lookahead bias. Returns are excess over SPY across the identical window.

HORIZON	EVENTS	MEAN EXCESS	MEDIAN EXCESS	HIT RATE
5 trading days	799	+0.09%	−0.40%	45.2%
21 trading days	669	+0.74%	−0.37%	49.0%
63 trading days	668	+0.42%	+0.08%	50.1%

Hit rates are a coin flip and the median trade is flat to negative. The positive means come from a handful of large winners, not a broad edge. **That is the argument for the desk, not against it** — if the average disclosed purchase is noise, the value has to come from selection.

Every limit is code, not a prompt

```
src/capitoldesk/config.py Every limit is a number in a file, not a prompt

class RiskLimits:

    max_notional_per_trade: float = 5_000.0
    max_notional_per_day: float = 25_000.0
    max_open_positions: int = 25
    max_contracts_per_order: int = 20
    max_pct_of_open_interest: float = 0.05 class="c"># don't be >5% of a contract's OI
    min_open_interest: int = 50 class="c"># skip illiquid contracts
    max_spread_pct: float = 0.20 class="c"># skip if bid/ask spread > 20% of mid
    min_days_to_expiry: int = 21
    max_days_to_expiry: int = 400
    max_filing_age_days: int = 75
```

Marketable limit orders only — never a market order on an option. And the desk refuses to start against a live endpoint.

CAUGHT IN DEVELOPMENT

Order reconciliation began writing real broker statuses while every limit query still filtered on 'placed'. The daily cap and position limit **silently stopped binding**.

Found, fixed, and pinned with parametrised regression tests asserting that live statuses consume the caps and dead ones release them.

45 tests cover the deterministic machinery — sizing under caps, spread validation, option-symbol parsing, the paper-only guard.

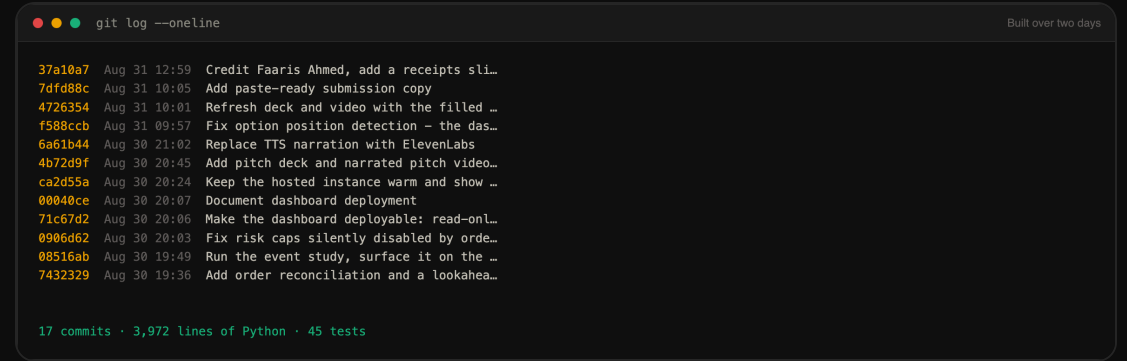
```
pytest tests/ -q Deterministic machinery under test

..... [100%]
./venv/lib/python3.12/site-packages/websockets/legacy/__init__.py:6
/Users/faaris/WORKSPACE/team-shaw-hackathon/.venv/lib/python3.12/site-packages/websockets/legacy/__init__.py:6: DeprecationWarning: websockets.leg
```

The model can argue for a trade. It cannot argue for a bigger one.

What broke, and how we knew

- **A one-sided quote halved every price.** The free feed returned `ask=0`, so a midpoint read AMAT at \$221 instead of \$461. Caught because the strategist *refused to trade* on incoherent option prices.
- **Extraction silently truncated.** A 274-transaction scan blew past the output cap and surfaced only as a JSON parse error. Now streams, and fails loudly.
- **Reconciliation disabled the risk caps.** Broker statuses stopped matching 'placed', so the daily cap silently stopped binding.
- **The dashboard showed an empty book.** In an OCC symbol, index -9 is the call/put letter — so the position filter matched nothing. Found the morning the orders filled.



```
git log --oneline

37a10a7 Aug 31 12:59 Credit Faaris Ahmed, add a receipts sli...
7dfd88c Aug 31 10:05 Add paste-ready submission copy
4726354 Aug 31 10:01 Refresh deck and video with the filled ...
f588ccb Aug 31 09:57 Fix option position detection - the das...
6a61b44 Aug 30 21:02 Replace TTS narration with ElevenLabs
4b72d9f Aug 30 20:45 Add pitch deck and narrated pitch video...
ca2d55a Aug 30 20:24 Keep the hosted instance warm and show ...
00040ce Aug 30 20:07 Document dashboard deployment
71c67d2 Aug 30 20:06 Make the dashboard deployable: read-onl...
0906d62 Aug 30 20:03 Fix risk caps silently disabled by orde...
08516ab Aug 30 19:49 Run the event study, surface it on the ...
7432329 Aug 30 19:36 Add order reconciliation and a lookahea...

17 commits · 3,972 lines of Python · 45 tests
```

Every one of these was found by running the thing against real market data — not by reading the code.

RECEIPTS

Not a mockup

Real filings, real orders, real fills — straight from the terminal.

desk extract 20035143

One filing, fully structured

\$ desk extract 20035143

Hon. Nancy Pelosi CA11 filed 2026-08-21

7 transactions extracted

P BE [ST] \$1,000,001-\$5,000,000 2026-07-24

P BE [OP] \$1,000,001-\$5,000,000 2026-07-24 -> 100x call \$100 exp 2027-06-17

P BE [ST] \$ 500,001-\$1,000,000 2026-07-28

P BE [OP] \$ 500,001-\$1,000,000 2026-07-28 -> 100x call \$100 exp 2027-06-17

P INTC [OP] \$ 250,001-\$500,000 2026-07-24 -> 50x call \$50 exp 2027-06-17

P INTC [ST] \$ 500,001-\$1,000,000 2026-07-24

P - [AB] \$ 500,001-\$1,000,000 2026-07-27

desk positions

Live book, marked against Alpaca

PA3K1WQNBURG equity \$99,574.87 cash \$92,889.87

Symbol	Qty	Avg cost	Current	Market value	P&L	P&L %
BE270617C00175000	1	\$79.55	\$75.00	\$7,500.00	\$-455.00	-5.7%
BE270617C00230000	-1	\$59.05	\$57.10	\$-5,710.00	\$195.00	+3.3%
INTC270115C00090000	2	\$14.50	\$13.95	\$2,790.00	\$-110.00	-3.8%
INTC270617C00090000	1	\$21.60	\$21.05	\$2,105.00	\$-55.00	-2.5%

RUNNING NOW

Live, autonomous, and auditable

Positions and P&L stream from a real Alpaca paper account. `desk` `loop` runs unattended — scanning for filings, holding them when the market is closed rather than pricing into a session it hasn't seen, and handing the book to a review agent over **Alpaca's MCP server**.

Capitol Desk

PAPER

PA3K1WQNBWVG

READ-ONLY DEMO

Mirrors public STOCK Act disclosures into options. Updated 09:59:55 EDT.

Positions and account are live from Alpaca. Scanning is disabled here — clone the repo and run `desk` `run` to watch it read filings and trade.

PORTFOLIO VALUE

\$99,614.87

▼ -385.13 today · \$92,890 options buying power

Filings read

44

Transactions parsed

650

Members tracked

36

Trades placed

3

Capital deployed

\$7,154

Passed over

635

OPEN POSITIONS

Contract	Qty	Avg cost	Last	Value	Unrealised	P&L
BE270617C00175000	1	\$79.55	\$76.45	\$7,645.00	-310.00 (-3.9%)	
BE270617C00230000	-1	\$59.05	\$59.50	\$-5,950.00	-45.00 (-0.8%)	
INTC270115C00090000	2	\$14.50	\$14.40	\$2,880.00	-20.00 (-0.7%)	
INTC270617C00000000	1	\$21.60	\$21.50	\$2,150.00	10.00 (0.5%)	

CAPITOL DESK

The interesting agent isn't the one that trades. It's the one that declines.

Live demo

capitol-desk.onrender.com

Source

[github.com/faarisaahmed/
team-shaw-hackathon](https://github.com/faarisaahmed/team-shaw-hackathon)

Built with

Claude Opus 5 · Claude Agent SDK
Alpaca MCP Server · Alpaca API

Faaris Ahmed